

The Diminishing Liquidity Premium

By

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Abstract

Previous evidence suggests that less liquid stocks entail higher average returns. This relation is explained by compensation that investors require for investing in illiquid assets. We present evidence using NYSE data that both the sensitivity of returns to illiquidity and illiquidity premia have significantly declined over the past four decades. Specifically, average annual illiquidity premia have declined from about 2% in the 60s and 70s to an amount that we cannot statistically distinguish from zero, starting in the mid 80s. Moreover, the profitability of trading strategies based on buying illiquid stocks and selling liquid stocks has declined over the past four decades rendering such strategies virtually unprofitable. The annual Fama-French four factor alpha of a portfolio long in illiquid stocks and short in liquid stocks declined from an average of 8.6% between 1964 to 1973, to an amount that is not statistically significant, starting from the mid 80s. The results are robust to different measures of illiquidity, and are distinct from time trends in the size anomaly. We argue that the gradual introduction and proliferation of index funds and exchange traded funds is a possible explanation for these results.

1 Introduction

Starting from the seminal work of Amihud and Mendelson (1986), it has been argued that transaction costs and liquidity have an important effect on the prices of financial assets. Amihud and Mendelson assume exogenous trading frequencies and conclude that transaction costs are priced: less liquid securities entail higher expected returns, which in turn benefit investors with long trading horizons. Furthermore, the price effects of illiquidity may be first order since the price reflects the present value of all future transaction costs.¹

The empirical presence of illiquidity premia was investigated extensively for various asset classes and using several different methodologies.² Amihud and Mendelson (1986, 1989), Amihud (2002), Brennan and Subrahmanyam (1996), Brennan, Chordia, and Subrahmanyam (1998), and Eleswarapu (1997), among many others find that different measures of illiquidity are associated with higher future stock returns.³ Amihud and Mendelson (1991) find similar results in bond markets. Recent studies (e.g. Pastor and Stambaugh (2003), Acharya and Pedersen (2005), Sadka (2006), and Korajczyk and Sadka, (2007)) have taken liquidity premia one step further, and argued that liquidity is a priced risk factor.⁴

¹This idea has been subject to some theoretical debate as models that endogenize the trading frequency suggest that the price effect of transaction costs is of second order. See Constantinides (1986) and Vayanos (1998), as opposed to Jang, Koo, Liu, and Loewenstein (2007).

²For a comprehensive survey of this literature see Amihud, Mendelson, and Pedersen (2006).

³Some exceptions should be noted. Hasbrouck (2006) finds only weak pricing effects of illiquidity. Spiegel and Wang (2005) find that much of the pricing effect of illiquidity is subsumed by the effect of idiosyncratic risk. Barclay, Kandel, and Marx (1998) do not find a significant price effect following a change in liquidity resulting from stocks moving between Nasdaq, NYSE and AMEX.

⁴Kamara, Lou, and Sadka (2007) show divergence over time between the systematic liquidity component of small-cap and large-cap firms. They attribute their findings to patterns of institutional ownership over their sample period.

In this paper we argue that the sensitivity of returns to illiquidity has significantly declined over the past 40 years. Obviously, liquidity itself has improved over the years following numerous regulatory reforms (such as decimalization) and technological improvements. Our key insight, however, is distinct from this point. We have no prediction regarding illiquidity itself; rather, we argue that the effect of each unit of illiquidity on returns has declined over the years. In a practical sense, we hypothesize that if we put returns on the left hand side of a regression and common measures of illiquidity on the right hand side, then the coefficient of illiquidity declines over time, and the total effect (illiquidity times the coefficient) declines over time. We further hypothesize that the profitability of liquidity-based long-short trading strategies has decreased over the years.

We test these hypotheses on all NYSE common stocks between 1964 and 2005. Our main measure of illiquidity is an inflation adjusted version of Amihud's measure (Amihud, 2002), defined as the absolute return per unit of dollar volume. This measure is easy to calculate from daily CRSP data, and it gained popularity in recent years as a useful measure of illiquidity. Adjusting for inflation is needed to facilitate comparisons over time, since the real value of dollar volume has changed over the years. We also present robustness tests of our analysis using turnover and dollar volume as alternative measures of liquidity. Unlike the traditional measures of illiquidity (e.g. bid-ask spread and depth), these measures do not rely on intraday data, and are thus available for analysis of data prior to the early 1980s.

We first show that in a cross-sectional study, illiquidity is indeed associated with higher average returns – confirming prior studies. We then turn to a time-series approach.

We use a sequence of parametric and non-parametric tests to show that over the past 42 years, both the sensitivity of returns to illiquidity and illiquidity premia have declined. For example, we find that the sensitivity of monthly returns to illiquidity has dropped from being positive and large both statistically and economically between 1964 and 1973 to an amount that is not statistically different from zero starting from the mid 1980s. Furthermore, the average annual illiquidity premia have declined from about 2% in the 60s and 70s to an amount that is not significantly different from zero starting in the mid 80s.

We further show that typical trading strategies based on buying illiquid stocks and selling liquid stocks have lost much of their profitability over the years. Using an out-of-sample analysis we show that a trading strategy that buys the top decile of illiquid stocks and sells the bottom decile yielded an average annual Fama-French four-factor alpha of 8.6% between 1964 and 1973. Starting from the mid 80s, the average alpha of such strategies is not significantly different from zero.

Naturally, firm size and liquidity are highly correlated. Importantly, our results regarding the decline in the effect of illiquidity on returns are distinct from well documented trends in the small-firm anomaly (e.g. Fama and French (1992), Dichev (1998), and Schwert (2003)). First, in our regression analysis we control for size effects. Second, in our analysis of liquidity-based trading strategies we pre-sort the data by size. The decline in profitability of liquidity-based strategies applies to all size cohorts.

Overall, the results show a strong decline in the effect of illiquidity on expected returns. Starting from the mid 1980s there does not appear to be a liquidity premium, and liquidity-based trading strategies appear to be unprofitable.

These results are important for both valuation and asset management applications. The existence of liquidity premia plays a central role in the valuation of financial assets. A 1% increase in discount rates (which is a typical annual liquidity premium over our entire sample period) may easily translate into more than 10% decrease in valuation. Furthermore, liquidity-based long-short trading strategies have become very common over the years, especially for hedge funds. Our findings cast doubt on the profitability of such strategies in recent years (at least for stocks listed on NYSE).

We offer a possible explanation for these results stemming from financial innovation. Specifically, it may be that the presence of financial instruments that allow investors to buy and sell illiquid assets indirectly (such as index funds and ETFs) works to lower the sensitivity of returns to illiquidity. These instruments enable investors to hold illiquid stocks indirectly for very low transaction costs, reducing the sensitivity of returns to illiquidity.

The rest of the paper is organized as follows. Section 2 describes the data and the main variables of interest. Section 3 presents the empirical results regarding the decline in liquidity coefficients and liquidity premia. Section 4 presents the results regarding the decline in profitability of liquidity-based trading strategies. Section 5 presents robustness tests using alternative liquidity measures. Section 6 discusses the possible explanation. We conclude in Section 7.

2 Data and Main Variables of Interest

Our sample consists of all NYSE common stocks from CRSP between January 1964 and December 2005 with share codes 10 or 11 (common shares). Thus, we exclude

ETFs, REITs, closed end funds, primes, and scores.

We do not consider NASDAQ stocks for two reasons. First, our liquidity measures are a function of volume. It is well known that volume in NASDAQ is inflated due to inter-dealer trading (see Atkins and Dyl (1997)). Furthermore, the amount of inflation in volume numbers has varied over the years due to the introduction of ECNs and other mechanisms that enable traders to bypass the dealer system. This prevents the use of Amihud's measure in a time-series analysis of NASDAQ liquidity. The second reason is that CRSP data on NASDAQ until 1983 is very scarce, precluding a valid longitudinal analysis of NASDAQ illiquidity using *any* liquidity measure. The results reported in the paper also exclude AMEX stocks. Our concern here is that the reduction in the number of common stocks traded on AMEX over our sample period can generate trends in the data that might contaminate our findings. For example, starting from the year 2000, the number of stocks available for our analysis on AMEX is only about 200. We have repeated the entire analysis including AMEX data. None of the results changes in a material way.

Our main illiquidity measure is a modified version of the measure presented in Amihud (2002). This is an annual measure of illiquidity in the spirit of Kyle's (1985) lambda, calculated based on the annual averages of daily absolute price changes adjusted for dollar volume and for inflation. Formally, Amihud's measure for firm i in year t is denoted by $Illiq_{i,t}$ and is given by

$$Illiq_{i,t} = \frac{1}{D_{it}} \sum_{d=1}^{D_{it}} \frac{|R_{idt}|}{VOLD_{idt} \cdot \inf_{dt}},$$

where, R_{idt} is the return of stock i in day d of year t , $VOLD_{idt}$ is the dollar volume (in

millions) of stock i on day d of year t , D_{it} is the number of available trading days for stock i in year t , and inf_{dt} is an adjustment factor for inflation, which allows us to present the *Illiq* measure using end-of-2005 prices. Such an adjustment is necessary since the real economic meaning of dollar-volume has changed significantly over the years.

Following Amihud (2002), to ensure sufficient amount of data in the calculation we only calculate the illiquidity measure for stocks that satisfy the following two requirements: (i) The stock must have return data for at least 200 trading days during the year; and (ii) The stock must be listed at the end of the year, and the end year price is higher than \$5. We also censored the upper and lower 1% of the distribution of the illiquidity measure to avoid outliers. After accounting for these restrictions, the number of stocks in our sample ranges from 1,053 in 1964 to 1,304 in 2005.⁵ For some of the analysis we control for the book-to-market ratio. The available source of data for book (and book-to-market) values is Compustat, which does not cover the entire CRSP universe. The number of stocks in our analyses that employs the book-to-market ratio ranges from 754 in 1964 to 1,284 in 2005.

Table 1 presents summary statistics of *Illiq* (for firms with book-to-market data). The median *illiq* is 2.4% suggesting that the median absolute price reaction to \$1 million of volume (in 2005 prices) is 2.4%. The average *illiq* is 7.4%, indicating that this measure is quite skewed, presumably due to relatively small and illiquid stocks. Figure 1 plots the evolvement over time of *Illiq*. For each year the figure depicts the average cross-sectional illiquidity measure. While illiquidity seemed to fluctuate significantly during the 70s and 80s, it appears to be trending down since the early 90s. This is not a big surprise given the

⁵ These are average numbers. Our analysis is done on a monthly basis and the actual number changes somewhat from month to month.

several market reforms (such as decimalization) that took effect during these years. It is important to distinguish between this gradual improvement in liquidity and the effect of illiquidity on returns, which is the focus of this paper. The fact that liquidity has improved does not by itself mean that the sensitivity of returns to illiquidity has changed or that liquidity premia have gone down.

Following Kamara, Lou, and Sadka (2006) we also calculate the systematic component of illiquidity for each stock. We do this by first calculating a market-wide measure of daily illiquidity as the equal-weighted average of the *Illiq* measures for all available stocks in a specific day. We denote the market illiquidity measure for day d in year t by $Illiq_{M,dt}$. The systematic illiquidity component of stock i in year t (also referred to as $Illiqbeta_{it}$) is derived from the following time series regression

$$Illiq_{idt} = \alpha + \beta_{ILLIQ,it} Illiq_{M,dt} + \varepsilon_{idt}, \quad (1)$$

which we estimate using all available daily observations for stock i during year t . Table 1 shows summary statistics for the sub-sample of stocks that have book-to-market data. The average of $Illiqbeta$ over our sample period is 0.95, and the median is 0.17. Again, this variable is highly skewed. While most stocks have just low sensitivity to market-wide liquidity, a relatively small number of stocks have very high sensitivity.

Table 1 also reports summary statistics for additional variables used in the analysis. First, following Amihud (2002), the beta is calculated using a methodology in the spirit of Fama and French (1992) as follows. At the end of each year t stocks are ranked by their market capitalization and divided into ten equal portfolios. We then estimate ten market models for year t in which the dependent variable is the daily excess return of one of the above ten portfolios, and the explanatory variable is the daily market excess return

using the Scholes and Williams (1977) method. The beta assigned to each individual stock i in year t is the estimated beta of the size portfolio containing this stock. The resulting average beta in the sample is 0.97, and the median is 0.98.

The main dependent variable is the stock return. We use monthly returns from CRSP. Following Shumway (1997) we adjust the returns to account for delisting bias.⁶ Other variables of interest whose summary statistics are reported in Table 1 are market capitalization (*Size*), the book-to-market ratio (*BM*), the daily standard deviation of returns (*Sdret*), the dividend yield (*Divyld*), and two variables that account for past returns in an attempt to capture momentum effects: *R100* is the accumulated return over the last 100 days of the year; and *R100yr* is the accumulated return from the beginning of the year until the last 100 days.

Table 2 reports averages of the monthly cross-sectional correlations between the main variables of interest from January 1964 to December 2005. Note the high negative correlation between *Lsize* (the log of *Size*) and *Illiq* (-0.61). As expected, small stocks are more illiquid. *Illiq* is also highly correlated with *Illiqbeta*, suggesting that illiquid stocks are also strongly affected by market illiquidity. Finally, note the positive correlation between *Illiq* and the standard deviation of returns. This is consistent with Spiegel and Wang (2005).

Given the high correlation between liquidity and size, for some of our analysis we would like to separate the liquidity effect from any size effects. In those cases we replace

⁶ Our approach here is similar to Amihud (2002). The last return used is either the last return available on CRSP, or the delisting return, if available. Shumway (1997) finds an average delisting return of -30% using OTC returns of delisted stocks. We thus assign a return of -30% if a delisting is coded by 500 (reason unavailable), 520 (went to OTC), 551–573 and 580 (various reasons), 574 (bankruptcy) and 584 (does not meet exchange financial guidelines).

Lsize with the residuals obtained from the following cross-sectional regression estimated separately for each year t :

$$Lsize_{i,t} = \alpha_t + \beta_t Illiq_{i,t} + \varepsilon_{i,t}. \quad (2)$$

We denote the residuals from this model by *Reslsize* _{i,t} . By construction these residuals capture the component of the size effect which is orthogonal to the illiquidity effect.

3 Analysis of Time Trends

Our main hypotheses are that (i) the sensitivity of returns to illiquidity has declined over the past 40 years; and (ii) liquidity premia have declined over this time period. We test these hypotheses by employing the Fama-MacBeth (1973) methodology to estimate the sensitivity of returns to illiquidity from monthly cross-sections of stocks, and use a series of parametric and non-parametric tests to check for a trend in the liquidity coefficients and liquidity premia.

For each month m in year t between January 1964 and December 2005 (504 months) we estimate a cross-sectional regression of the form:

$$R_{imt} = \alpha_{mt} + \sum_{j=1}^J \beta_{jmt} X_{ij,t-1} + \varepsilon_{imt}. \quad (3)$$

That is, we regress the returns of firm i in month m of year t on a set of J explanatory variables calculated using data from year $t-1$. This ensures that the explanatory variables are known to investors at the time that monthly returns are being set. The main explanatory variable is Amihud's illiquidity measure (*Illiq*). We also include additional explanatory variables, which have been shown (or are suspected) to be determinants of returns. These are: beta (*Beta*), illiquidity-beta (*Illiqbeta*), size (*Lsize* or *Reslsize*),

momentum ($R100$, and $R100yr$), standard deviation of returns ($Sdret$), and dividend yield ($Divyield$).

For the purpose of our analysis, the main output from the above regressions is twofold. First, we obtain 504 monthly estimates of the sensitivity of returns to illiquidity – one for each month in the sample period. Second, by multiplying these monthly coefficients with the average illiquidity measure of the relevant month we obtain an estimate of the monthly illiquidity premium. We then use these coefficient estimates and the estimates of the illiquidity premia to test for trends.

To begin, Table 3 presents a standard Fama-MacBeth analysis of the entire sample period. For each explanatory variable the table reports the average of the coefficient from all of the monthly observations, as well as a t-statistic testing against the null hypothesis that this average is zero. The results are quite typical to this kind of tests. First, as noted by Fama and French (1992), *beta* is not priced. Second, *Lsize*, book-to-market, and momentum are all priced in the familiar way: smaller stocks entail higher returns, “value” (high book-to-market) stocks entail higher returns, and winners stay winners in the short-run.

The relevant part for our purposes is the coefficient of the illiquidity measure. It is significant and positive in six out of the eight specifications. To see the economic magnitude of the coefficients consider for example specification (1). A coefficient of 0.89 implies that an increase of one standard deviation in *Illiq* (0.13. see Table 1) would increase monthly returns by $0.89\% \times 0.13 = 0.116\%$. On a yearly basis this amounts to approximately 1.4%. The average monthly illiquidity premium estimated with this specification (bottom of the table) is 0.08%, which translates into an annual premium of

approximately 0.96%. Note that these numbers are economically large. A 1% increase in discount rates would typically translate in standard valuation models into a discount of about 10%-20% in value (depending on the discount rate and the assumptions on expected growth).

It seems then that over the entire sample period, illiquidity is priced in an economically and statistically significant way in most specifications, but not in all of them. We argue that the somewhat inconclusive significance of the illiquidity coefficients in Table 3 reflects a mix of highly significant illiquidity coefficients until the mid 80s and low and insignificant illiquidity coefficients in the period starting in the mid 80s.

To get a first impression of the plausibility of this assertion, Figure 2A plots the illiquidity coefficients obtained from Eq. (3) (under specification 1) over time. For each year the figure depicts the average monthly estimates of the illiquidity coefficient for that year. While these coefficient estimates are noisy and seem to fluctuate quite a bit, there appears to be a clear downward trend. Similarly, Figure 2B plots the average illiquidity premia, which again appear to be quite noisy but clearly downward trending. The analysis below formally tests for the existence of these trends.

In our first attempt to identify trends in the illiquidity coefficients we divide the 42 years in our sample period into four sub-periods of 10 or 11 years. The idea in slicing the entire sample period into sub-periods is to neutralize some of the noise in the monthly coefficients by averaging them over several years of data. For example, by examining Figure 2A one might wonder whether the trend is an artifact of the very high coefficient in 1969, or the very low coefficient in 1999. By using 10-11 years of data in each sub-period we have 120-132 monthly observations per sub-period, which is likely to alleviate

some of the inevitable noise. Another approach would be to use non-parametric tests for trend (see below).

The sub-periods we consider are as follows. Period 1 is 1964-1973, Period 2 is 1974-1984, Period 3 is 1985-1995, and Period 4 is 1996-2005. We then apply the Fama-MacBeth analysis (as above) separately to each of the sub-periods and compare the resulting coefficients.

The results are presented in Table 4. As before, the coefficients reported in the table are the averages of the coefficients of the monthly cross-sectional regressions over the relevant time periods. The two specifications differ in the way they control for size – specification (1) uses the component of size that is orthogonal to illiquidity, whereas specification (2) uses the size variable itself. Consider specification (1). The average illiquidity coefficients in Periods 1 and 2 are 3.14 and 0.99 with t-stats 2.80 and 2.00 respectively. By contrast, the Illiq coefficients in Periods 3 and 4 are not significantly different from zero. The average monthly illiquidity premium period 1 is 0.19%, or approximately 2.3% in annual terms. In period 2 the monthly illiquidity premium is lower at 0.14% (about 1.7% annually). By contrast, in periods 3 and 4 the monthly illiquidity premium is not significantly different from zero. Similar conclusions follow from specification (2), only that the liquidity effect vanishes even sooner during Period 2.

Given these results, and to further increase the power of our tests by decreasing noise, we group periods 1 and 2 into one early sub-period extending from 1964 to 1984, and periods 3 and 4 into one later sub-period extending from 1985 to 2005. We then repeat the analysis for each of the two half-sample periods. The results are reported in Table 5. As before, the two specifications differ by the way we control for size.

Consider specification (1). Between 1964 and 1984 the *Illi*q coefficient estimate is 2.01 and the monthly illiquidity premium is 0.17%. Both are highly significant. By contrast, between 1985 and 2005 both the illiquidity coefficient and the illiquidity premium are not statistically different from zero. A similar story is told in specification (2). A comparison of the coefficient and liquidity premia across the two periods (bottom of table) shows a statistically significant decline in both specifications.

We next turn to a set of non-parametric tests of trend. The main concern from the above parametric analysis is that the averaging of monthly coefficients across sub-periods does not do a good enough job in cancelling-out noisy observations, and that the results are driven by just a few outliers. The advantage of non-parametric tests of trend is that they only consider the ordering of the magnitudes of the coefficients, and do not account for the magnitude itself. Thus, extremely high or low coefficient estimates do not have an unusual effect on the test results, as opposed to parametric methods.

We apply two standard non-parametric tests of trend: the Spearman test and the Kendall test. Similar to the standard Pearson's correlation, these non-parametric tests produce a number between -1 and 1, and allow for rejecting the null hypothesis of no trend. We apply these tests to the time series of 504 illiquidity coefficients and similarly to the time series of estimated monthly liquidity premia. The non-parametric tests as well as standard Pearson's correlations are reported in Table 6. The results show a significant negative time trend in both the liquidity coefficients and illiquidity premia. The p-values are less than 1% in all of the specifications, again supporting our hypotheses.

Overall, the parametric and non-parametric analyses reinforce one another. They both support the claim that liquidity coefficients and illiquidity premia have been trending down over the years.

4 Profitability of Liquidity Based Trading Strategies

The higher expected returns of illiquid stocks have long attracted long-term investors, who try to reap the higher gains, not having to liquidate early. Anecdotal evidence suggests that many hedge-funds use long-short strategies buying illiquid stocks and short-selling liquid stocks of the same class. Are such strategies still profitable given our results on the decline in liquidity premia?

To answer this question, for each year between 1964 and 2005 we sort the stocks into deciles based on their annual Amihud illiquidity level. The most illiquid stocks (highest Amihud measure) in each year are placed in Decile 1 (the top decile), whereas the most liquid stocks are in Decile 10 (the bottom decile). We then construct three rolling portfolios. The first is a portfolio of the most illiquid stocks based on illiquidity estimations in the previous year. Thus, this portfolio assigns equal weights to all stocks in the top liquidity decile for a given year, and it rebalances the holdings once a year on January 1st. The second portfolio is composed of the most liquid stocks, and thus assigns equal weights to all the stocks in the bottom decile, again rebalanced annually. A third portfolio is a long-short portfolio that is long in the illiquid portfolio and short in the liquid one.

Panel A of Table 7 reports the average monthly excess returns for these three portfolios over the four sub-periods described above. The most striking result in this panel is the excess returns on the long-short portfolio. The average monthly excess return on this portfolio between 1964 and 1973 is 0.73%, which amounts to approximately 8.8% annually. Surprisingly, the average excess return of this long-short portfolio is even higher (1.47%) in the second sub-period. However, in the two sub-periods starting from 1985, the average excess return of the long-short portfolio is not statistically different from zero. To test for trend, in Panel B we compare the sub-periods 1964-1984 to 1985-2005. The difference in excess returns of the long-short strategy between the two sub-periods is highly significant (t-stat of 3.40).

To further evaluate the profitability of liquidity-based strategies we estimate out-of-sample alphas of the three portfolios relative to the Fama-French four-factors (excess market return, HML, SMB, and UMD). Our approach here is similar to the one taken in Brennan, Chordia and Subrahmanyam (1998), Chordia, Subrahmanyam, and Anshuman (2001), and Pastor, Stambaugh (2003). Specifically, for each month m between 1964 and 2005 we regress the monthly excess returns of the three liquidity portfolios on the returns of the Fama-French four factors during the preceding 60 months: $m-60$ to $m-1$. Thus, for each month m in our sample period we obtain an estimate of the 4-factor loadings as of that month. Denote these factor loadings by $\beta_{MKT,p,m}$, $\beta_{HML,p,m}$, $\beta_{SMB,p,m}$, and $\beta_{UMD,p,m}$, where, for example, $\beta_{MKT,p,m}$ stands for the loading on the market factor related to month m and portfolio p (one of the three liquidity portfolios). Now, for each month m we calculate the out-of-sample 4-factor alpha of portfolio p (denote $Alpha_{m,p}$) as the realized

excess return of the portfolio less the expected return calculated from the realized returns on the factors and the estimated factor loadings:

$$\begin{aligned} \text{Alpha}_{m,p} = & (RET_{p,m} - Rf_m) - \beta_{MKT,p,m} (RET_{MKT,m} - Rf_m) - \beta_{SMB,p,m} SMB_m \\ & - \beta_{HML,p,m} HML_m - \beta_{UMD,p,m} UMD_m, \end{aligned} \quad (4)$$

where $RET_{p,m}$, $RET_{MKT,m}$, and Rf_m are the realized returns on portfolio p , the CRSP value-weighted index, and the risk free rate, respectively during month m ; and SMB_m , HML_m , and UMD_m are the appropriate realized returns on the factor portfolios in month m .

For each of the three portfolios we thus obtain a time series of between 120 to 132 out-of-sample alpha estimates for each of the sub-periods defined above. Panel A of Table 8 reports the averages of these alpha estimates. Consider first the top decile portfolio. In the first two sub-periods a long investment in this portfolio yielded an average out-of-sample monthly four-factor alpha of 0.44% (about 5.3% in annual terms). In contrast, the average alpha of this portfolio is not significantly different from zero in the two later sub-periods. As for the bottom decile (the most liquid stocks) – a naked short position in these stocks yielded an average positive monthly alpha of 0.28% (3.4% in annual terms) in the sub-period 1. However, starting from period 2 and on, the average alpha on this portfolio is not significantly different from zero. Finally, the long-short portfolio tells a similar story. A strategy that is long in the most illiquid stocks and short in the most liquid stocks yielded average monthly alphas of 0.72% and 0.37% in the Periods 1 and 2 respectively (about 8.6% and 4.4% in annual terms). However, in the later periods, the average alphas are not significantly different from zero.

In Panel B we again divide the sample period to two sub-periods and compare the alpha estimates between these sub-periods. The results confirm a statistically and

economically significant decline in the profitability of the long, short, and long-short strategies. These strategies appear to be unprofitable starting from the mid 1980s.

To check the robustness of the results regarding the trends in excess returns and alphas of the long-short portfolio we have used the Spearman and Kendall non-parametric tests applied to the time series of all 504 monthly estimates. Figure 3 presents an illustration of the evolvement over time of the four-factor alphas. The Spearman and Kendall coefficients for the excess returns are -0.092 and -0.063 with p-values of 0.04 and 0.04 respectively. The Spearman and Kendall coefficients for the monthly alphas are -0.133 and -0.091 with p-values of 0.00 and 0.00 respectively. Again, these results reject the null hypothesis of no trend.

We next turn to examine more closely the important relation between illiquidity and firm size. Prior research has pointed out that the effect of firm size on expected returns has declined since the early 1980s (e.g. Fama and French (1992), Dichev (1998) and Schwert (2003)). Our Fama-MacBeth regressions control for firm size, already taking any trends in this variable into account. In our portfolio analysis we have taken into account the sensitivity of our liquidity-based portfolios to systematic size effects by adjusting for the SMB factor. Still, a concern is that the results obtained so far are somehow reflecting a size effect in a non-linear fashion.

To check whether this is indeed the case, we repeat the prior analysis by first sorting stocks by firm-size. That is, for each month in our sample period we sort the stocks into five size quintiles. We then construct five liquidity-based long-short portfolios as above within each size quintile. The portfolios are rebalanced annually. The idea is that if size effects are driving the decline in the profitability of liquidity-based portfolios,

then sorting first by firm size would eliminate the liquidity effect within each size quintile. By contrast, if we do see a decline in the profitability of liquidity-based portfolios within the size quintiles, then this effect is independent of documented trends in the size effect.

Table 9 reports separate four-factor alpha averages for liquidity-based long-short portfolios within each size quintile. Sizes 1-5 refer to the smallest to largest size quintiles respectively. The results confirm that the decline in the liquidity effect applies to all firm sizes. Panel A shows that in the first period (1964-1973), the average alphas are positive and significant both statistically and economically within all size quintiles. By contrast, the average alphas are not significantly different from zero for all size quintile in the last period (1996-2005). The size quintiles do differ to some extent in the path of the decline in alphas. Panel B divides the sample period to two sub-periods. For all size quintiles the alphas are positive and significant in the first sub-period, and are not significantly different from zero in the second sub-period. To further perform a formal test regarding the difference between the two periods we constructed a “grand portfolio” that has equal weights in each one of the five long-short portfolios. In contrast to the portfolios in Table 8, this portfolio is forced to have stocks from all size quintiles. We then estimated the average alphas on this “grand portfolio” in both sub-periods. The average alpha monthly alpha in the first sub-period is 0.47% with a t-stat of 4.17, whereas in the second sub-period the average alpha is 0.07% but insignificant. The difference between the two is significant with a t-stat of 2.63.

Overall, these results confirm that the decline in the profitability of liquidity-based portfolio is not related to time trends in the size effect, and it applies to all size levels.

5 Robustness –Alternative Liquidity Measures

Our focus throughout the paper was on one illiquidity measure suggested by Amihud (2002). This measure has been used for example in Acharya and Pedersen (2005), Korajczyk and Sadka (2007), and Sadka and Kamara (2007). The ease of calculation for periods before the availability of intraday data, the clear economic content, and the recent popularity of this measure led us to choose it as our primary measure. However, it is important to check whether the same trends are observed when other liquidity measures are used.

Traditional intra-day liquidity measures such as bid-ask spread, and market depth are not available prior to the early 1980s. This prevents the use of these measures for a longitudinal study. In this section we present robustness results using two additional common measures of liquidity that are available for a long-term analysis: average daily dollar volume and annual turnover (annual share volume divided by number of outstanding shares). These measures have been used as liquidity proxies by Bernnan, Chordia, and Subrahmanyam (1998), Chordia, Subrahmanyam, and Anshuman (2001), Korajczyk and Sadka (2007), and Datar, Naik, and Radcliffe (1998), among many others.

Panel A of Table 10 presents summary statistics for turnover and dollar volume during our sample period. The median yearly turnover in the sample is 43%, and the median daily dollar volume is \$403 million (exponent of 6.01). Panel B reports the correlations between the three liquidity measures used in this paper as well as their

correlation with firm size. It is important to note that while Amihud's measure is a proxy for illiquidity, both dollar volume and turnover are positive measures of liquidity. This explains the positive correlation between turnover and volume and the negative correlation between these two and Amihud's measure. Note that turnover is almost completely orthogonal to size, making it an appealing proxy for analyzing liquidity effects that are independent of size.

We have repeated our entire analysis using these two additional liquidity measures. The qualitative results are very similar to those obtained using Amihud's measure, and in some occasions show an even stronger decline in the effect of liquidity on returns. For brevity, we report here just selective results using these measures.

In Table 11 we present the results from Fama-MacBeth regressions similar to those presented in Tables 5 using the alternative liquidity measures. Specification 1 uses turnover whereas specification 2 uses dollar volume. Being positive measures of liquidity, we expect the coefficients of both turnover and dollar volume to be negative and to trend toward zero (decline in absolute value) over time. The results show a significant decline in the absolute value of the coefficients for both liquidity measures. For example, the coefficient of turnover is -0.82 (with t-stat -4.18) in the first period, and is not significantly different from zero in the second period. Similarly, the coefficient of dollar volume is significant and negative until the mid-80s and is not significantly different from zero from that time on. These results are consistent with our findings using Amihud's measure.

Table 12 repeats the analysis in Table 9 using the alternative liquidity measures for two sub-periods. As before, in each month we first pre-sort by size and then construct

illiquidity-based long-short portfolios for each size quintile. We perform this exercise twice: once using turnover and once using dollar volume. The results show a decline in the size of the four-factor alphas for both liquidity measures and over all size quintiles. As before, to obtain a statistic determining whether the decline in alphas is significant, we construct a “grand portfolio” assigning equal weights to the each one of the five size-based portfolios. We then compare the average alphas in the two sub-periods. The comparison shows that average alphas declined for both the turnover and the dollar volume measures (T-stats of 1.84 and 1.89 respectively).

Overall, our robustness tests indicate that the decline in the sensitivity of returns to illiquidity is not unique to the Amihud’s measure of illiquidity.

6 Possible Explanation and Discussion

The empirical results presented so far beg for an explanation. Why would the sensitivity of returns to illiquidity and liquidity premia decline over the years? Our explanation relies on innovation in financial markets. Specifically, we believe that the presence of financial instruments that allow investors to buy and sell illiquid assets indirectly (such as index funds and ETFs) works to lower the sensitivity of returns to illiquidity.

These instruments allow investors to hold illiquid stocks indirectly for very low transaction costs (similar to Cherkes, Sagi, and Stanton (2006) in the context of closed end funds). For example, direct investment in Russell 2000 stocks is quite expensive in terms of transaction costs. However, Russell 2000 ETFs (e.g. IWM) are highly liquid, presumably because (as in Subrahmanyam, 1991) there is almost no information trading

in ETFs.⁷ The ETFs and index funds themselves are long-term holders of the illiquid stocks, and thus incur only low transaction costs over the long-run. These low transaction costs cover the trading costs of the index funds and the ETFs since their trading frequency is very low. They employ a passive trading strategy, and trade only following index changes or as a result of significant mismatches between inflows and outflows. While these instruments charge management fees that can be avoided by direct investment in the underlying stocks, these management fees are typically very low.⁸

If investors can bypass the high transaction costs, then they should not be compensated in terms of returns. Thus, we expect that investors' compensation for investing in illiquid stocks will decline over the years as mutual funds, index funds, and ETFs become more popular. Importantly, none of the existing theoretical papers suggesting the existence of illiquidity premia considers investors that are allowed to invest in illiquid stocks indirectly through liquid funds that specialize in such stocks.

The past four decades have seen the introduction and proliferation of many such investment tools. Mutual funds have grown dramatically in the late 60s, index-funds have been introduced in the mid 70s, and ETFs have been introduced in the 90s. The coverage and popularity of these instruments has constantly increased over the years. Based on the argument above, we hypothesize that these tools have contributed to a decline in the sensitivity of returns to individual stocks' illiquidity, and to a decline in the liquidity premium.

Of course, this is not a complete formal theory. Building a formal equilibrium

⁷ For example, during April 2006 the average relative bid-ask spread of IWM was on average 31 times smaller than the average relative bid-ask spread of the shares composing the index : 0.018% vs. 0.558%.

⁸ For example, in 2006, the annual expense ratio of the Russell 2000 index fund of E*TRADE was 0.22%, and the expense ratio of IWM was 0.2%.

model that allows investors to choose between direct investment in assets and indirect investment through funds; and incorporating both transaction costs and equilibrium risk premia into the model seems a challenging task which is outside the scope of this paper. Therefore we do not claim that liquidity premia should completely vanish. Rather we offer a plausible explanation for their decline. Furthermore, our explanation cannot be easily tested: index funds and ETFs were introduced and gained popularity slowly over the years. Thus, it is not possible to identify one abrupt structural change that induced a decline in liquidity premia, ruling out an event study approach. Still, we view our explanation as one economically plausible story, which is consistent with the empirical results.

7 Conclusion

Using all NYSE common-stock data between 1964 and 2005 we find that both the sensitivity of returns to illiquidity and illiquidity premia have significantly declined over the past four decades. In fact, starting from the mid 80s we do not identify a significant liquidity premium. The profitability of liquidity-based trading strategies has declined significantly, and the results apply to all firm sizes, and thus, are orthogonal to trends in the size effect.

We suggest a possible explanation for these results: the presence of financial instruments such as index funds and ETFs that invest in illiquid stocks allow investors to bypass the high transaction costs associated with the direct trading of these stocks.

The results have important implications to valuation and asset management. A reduction of the average annual liquidity premium from 2% to 0% implies a very large price effect (depending on the discount rate and the expected dividend growth). Therefore our results seem to be related to the conclusion of Dimson, Marsh, and Staunton (2003) that a part of the realized equity returns in the second half of the 20th century is due to a reduction in the equity discount rates. Our findings suggest that a portion of this reduction may have been due to the decline of the liquidity components in expected returns. On the asset management side the result raises a question regarding the profitability of liquidity-based strategies (at least for NYSE stocks), which have become very popular in the years since Amihud and Mendelson's (1986) seminal paper.

Finally, it has been argued that other attributes of firms such as disclosure policy may affect firm's cost of capital (and value) through its effect on liquidity. These claims should be evaluated in light of our findings.

References

- Acharya, V. V. and L. H. Pedersen, 2005, Asset pricing with liquidity risk, *Journal of Financial Economics* 77, 375–410.
- Amihud, Y., 2002, Illiquidity and stock returns: Cross-section and time series effects, *Journal of Financial Markets* 5, 31–56.
- Amihud, Y. and H. Mendelson, 1986, Asset pricing and the bid-ask spread, *Journal of Financial Economics* 17, 223–249.
- Amihud, Y. and H. Mendelson, 1989, The effects of beta, bid-ask spread, residual risk and size on stock returns, *Journal of Finance* 44, 479–486.
- Amihud, Y. and H. Mendelson, 1991, Liquidity, maturity and the yields on U.S. government securities, *Journal of Finance* 46, 1411–1426.
- Amihud Y., Mendelson H., and L. H. Pedersen, 2006, Liquidity and Asset Prices, *Foundations and Trends in Finance* 1, 269–364.
- Atkins, A., and E. Dyl, 1997, Market Structure and Reported Trading Volume: Nasdaq Versus the NYSE, *Journal of Financial Research* 20, 291–304.
- Barclay, M. J. , E. Kandel, and L. M. Marx, 1998, The Effects of Transaction Costs on Stock Prices and Trading Volume, *Journal of Financial Intermediation* 7, 130–150.
- Brennan, M. J., T. Chordia, and A. Subrahmanyam ,1998, Alternative factor specifications, security characteristics, and the cross-section of expected stock returns, *Journal of Financial Economics* 49, 345–373.
- Brennan, M. J. and A. Subrahmanyam ,1996, Market microstructure and asset pricing: On the compensation for illiquidity in stock returns, *Journal of Financial Economics* 41, 441–464.
- Cherkes M., J. Sagi, and R. Stanton, 2006, A Liquidity-Based Theory of Closed-End Funds, *Review of Financial Studies*, forthcoming.

- Chordia, T., A. Subrahmanyam, and R. V. Anshuman, 2001, Trading Activity and Expected Stock Returns, *Journal of Financial Economics* 59, 3-32.
- Constantinides, G. M. ,1986, Capital market equilibrium with transaction costs, *Journal of Political Economy*, 94, 842–862.
- Datar, V. T., N. Y. Naik, and R. Radcliffe, 1998, Liquidity and Stock Returns: An Alternative Test, *Journal of Financial Markets* 1, 205-219.
- Dichev, I. D., 1998, Is the Risk of Bankruptcy a Systematic Risk? *Journal of Finance* 53, 1131-1147.
- Dimson, Elroy, Paul Marsh and Mike Staunton, 2003, Global evidence on the equity risk premium, *The Journal of Applied Corporate Finance*, Vol. 15, 8-19.
- Eleswarapu, V. R. 1997, Cost of transacting and expected returns in the Nasdaq market, *Journal of Finance* 52, 2113–2127.
- Fama, E.F., and K. R. French, 1992, The cross section of expected stock returns, *Journal of Finance* 47, 427–465.
- Hasbrouck, Joel, 2006, Trading Costs and Returns for US Equities: Estimating Effective Costs from Daily Data, working paper, NYU.
- Jang, B., H. K. Koo, H. Liu, and M. Loewenstein, 2007, Liquidity Premia and Transaction Costs, *Journal of Finance* 62(5), 2329–2366.
- Kamara, A., X. Lou, and R. Sadka, 2007, The Divergence of Systematic Liquidity and the Cross-Section of Stock Returns, Working Paper, University of Washington.
- Korajczyk R. A. and R. Sadka, 2007, Pricing the Commonality across Alternative Measures of Liquidity, *Journal of Financial Economics*, forthcoming.
- Loderer, Claudio and Lukas Roth, 2005, The pricing discount for limited liquidity: evidence from SWX Swiss Exchange and the Nasdaq , *Journal of Empirical Finance*, 12(2), 239-268.

- Pastor, L. and R. Stambaugh ,2003, Liquidity risk and expected stock returns, *Journal of Political Economy* 111, 642–685.
- Petersen M.A., 2007, Estimating Standard Errors in Finance Panel Data Sets: Comparing Approaches, Working Paper, Northwestern University.
- Roll, R., A, 1984, Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market, *Journal of Finance*, 39(4), 1127-1139.
- Sadka R., 2006, Momentum and post-earnings-announcement drift anomalies: The role of liquidity risk, *Journal of Financial Economics* 80, 309-349.
- Shumway, T., 1997, The delisting bias in CRSP data, *Journal of Finance* 52, 327–340.
- Spiegel, M. and X. Wang ,2005, Cross-sectional variation in stock returns: Liquidity and idiosyncratic risk, Working Paper, SSRN # 709781
- Subrahmanyam, A., 1991, A Theory of Trading in Stock Index Futures, *Review of Financial Studies* 4, 17-51.
- Schwert G. W., 2003, Anomalies and Market Efficiency, *Handbook of the Economics of Finance*, Edited by G.M. Constantinides, M. Harris and R. Stulz, Elsevier Science B.V.
- Vayanos, D. (1998), Transaction costs and asset prices: A dynamic equilibrium model, *Review of Financial Studies* 11, 1–58.

Table 1: Summary Statistics

The table reports the average of the monthly cross-sectional sample statistics for all the stocks in our sample that have book-to-market data. For each month in the sample we calculate the mean, median, min, max, and standard deviation. The table reports the time averages of these statistics. The sample period for the returns is 1964-2005. The sample period for the rest of the variables (explanatory variables) is 1963-2004. Ret is the CRSP monthly return adjusted for delistings. Illiq is Amihud's (2002) illiquidity measure adjusted for inflation presented in December 2005 prices. Illiqbeta is estimated as in Eq. (1). Beta is the size sorted portfolio beta calculated each year based on daily observations using the Scholes-Williams (1977) method. Sdret is the standard deviation of the daily returns. Divyld is the dividend yield. BM is the book-to-market ratio. R100 is the accumulated return over the last 100 days of the year. R100yr is the accumulated return from the beginning of the year until the last 100 days. Size is the end of year market capitalization.

Variables	Mean	Median	Min	Max	Std.
Ret (%)	1.20	0.68	-60.21	113.55	10.26
Illiq	0.074	0.024	0.001	1.01	0.13
Illiqbeta	0.95	0.17	-4.14	42.77	2.84
Beta	0.97	0.98	0.77	1.11	0.10
Sdret (%)	2.11	1.98	0.68	6.63	0.75
Divyld (%)	3.74	3.27	0.09	92.93	4.57
BM	0.77	0.70	-2.13	7.80	0.57
R100 (%)	5.49	3.77	-51.48	158.51	19.55
R100yr (%)	11.21	7.76	-63.03	234.31	27.40
Size (\$ million)	1,886	559	16	47,482	4,081
Obs	1,156	1,185	752	1,528	163

Table 2: Cross-Sectional Correlations

The table presents the average of the monthly cross-sectional Pearson's correlations from January 1964 to December 2005. Ret is the CRSP monthly return adjusted for stock delisting. Illiq is Amihud's (2002) illiquidity measure adjusted for inflation presented in December 2005 prices. Illiqbeta is estimated as in Eq. (1). Beta is the size sorted portfolio beta calculated each year based on daily observations using the Scholes-Williams (1977) method. Sdret is the standard deviation of returns calculated each year based on daily observations. Divyld is the dividend yield. LnBM is the logarithm of book-to-market ratio. R100 is the accumulated return over the last 100 days of the year. R100yr is the accumulated return from the beginning of the year until the last 100 days. Lnsiz is the logarithm of the end of year market capitalization (in thousands of dollars).

Variables	Illiq	Illiqbeta	Beta	Sdret	LnBM	Lnsiz	Divyld	R100	R100yr
Ret	0.017	0.014	0.009	-0.011	0.018	-0.014	0.009	0.019	0.009
Illiq		0.79	-0.10	0.19	0.20	-0.61	-0.04	-0.01	0.00
Illiqbeta			-0.05	0.22	0.14	-0.45	-0.07	0.01	0.02
Beta				0.07	-0.07	-0.01	0.00	0.02	0.02
Sdret					-0.04	-0.37	-0.41	0.00	0.09
LnBM						-0.28	0.19	-0.13	-0.13
Lnsiz							0.07	0.10	0.03
Divyld								-0.02	-0.08
R100									0.06

Table 3: Fama-MacBeth Regressions over the Entire Sample Period

The table presents the mean of the coefficients from monthly cross-sectional regressions of stock returns on the explanatory variables (Eq. (3)). The sample period is 1964-2005 resulting in 504 monthly regressions. Illiq is Amihud's (2002) illiquidity measure adjusted for inflation presented in December 2005 prices. Illiqbeta is estimated as in Eq. (1). Beta is the size sorted portfolio beta calculated each year based on daily observations using the Scholes-Williams (1977) method. Sdret is the standard deviation of daily returns. Divyld is the dividend yield. LnBM is the logarithm of book-to-market ratio. R100 is the stock accumulated return over the last 100 days of the year. R100yr is the accumulated return from the beginning of the year until the last 100 days. Lnsiz is the logarithm of the end year market capitalization (in thousands of dollars). Reslnsiz is the residual of Lnsiz calculated as in Eq. (2). RSQ is the average of the R-Squared of all the monthly regressions. The monthly illiquidity premium is the product of the monthly *Illiq* coefficient and the monthly average *Illiq*. T-stats are reported in small print below the coefficient estimates.

Variables	With LnBM				Without LnBM			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Intercept	1.02 3.04	1.01 3.03	2.02 3.38	2.00 3.36	0.92 2.78	0.92 2.75	2.24 3.73	2.22 3.71
Illiq	0.89 2.32	1.00 2.89	0.65 1.72	0.75 2.51	0.97 2.51	1.12 3.20	0.59 1.56	0.73 2.44
Reslnsiz	-0.08 -2.24	-0.08 -2.22			-0.10 -2.83	-0.10 -2.81		
Lnsiz			-0.08 -2.24	-0.08 -2.22			-0.10 -2.83	-0.10 -2.81
Illiqbeta	0.01 0.89		0.01 0.89		0.01 1.00		0.01 1.00	
Beta	0.44 1.28	0.43 1.27	0.44 1.28	0.43 1.27	0.38 1.14	0.38 1.13	0.38 1.14	0.38 1.13
Sdret	-0.20 -2.38	-0.20 -2.33	-0.20 -2.38	-0.20 -2.33	-0.19 -2.29	-0.19 -2.24	-0.19 -2.29	-0.19 -2.24
Divyld	-0.02 -1.88	-0.02 -1.86	-0.02 -1.88	-0.02 -1.86	-0.01 -0.72	-0.01 -0.70	-0.01 -0.72	-0.01 -0.70
R100	0.95 4.37	0.94 4.31	0.95 4.37	0.94 4.31	0.92 4.19	0.91 4.13	0.92 4.19	0.91 4.13
R100yr	0.40 3.28	0.39 3.15	0.40 3.28	0.39 3.15	0.36 2.92	0.34 2.79	0.36 2.92	0.34 2.79
LnBM	0.19 4.81	0.19 4.87	0.19 4.81	0.19 4.87				
RSQ	0.069	0.068	0.069	0.068	0.066	0.065	0.066	0.065
Liquidity Premium (%)	0.08	0.09	0.04	0.04	0.09	0.10	0.03	0.04
T-stat	3.35	3.95	1.63	2.46	3.67	4.32	1.58	2.49

Table 4: Fama-MacBeth Regressions - Four Sub-Periods

The table presents the mean of the coefficients from the monthly cross-sectional regression of stock returns on the explanatory variables (Eq. (3)). The sub-periods are: 1964-1973, 1974-1984, 1985-1995, and 1996-2005. *Illiq* is Amihud's (2002) illiquidity measure adjusted for inflation presented in December 2005 prices. *Illiqbeta* is estimated as in Eq. (1). *Beta* is the size sorted portfolio beta calculated every year based on daily observations using the Scholes-Williams (1977) method. *Sdret* is the standard deviation of the stocks returns calculated every year based on daily observations. *Divyld* is the dividend yield. *LnBM* is the logarithm of the book-to-market ratio. *R100* is the stock accumulated return over the last 100 days of the year. *R100yr* is the stock accumulated return from the beginning of the year till the last 100 days. *Lnsize* is the logarithm of the end year market capitalization (in thousands of dollars). *Reslnsize* is the residual of *Lnsize* calculated as in Eq. (2). *RSQ* is the average of the R-Squared of all the monthly regressions in a period. The monthly illiquidity premium is the product of the monthly *Illiq* coefficient and the monthly average *Illiq*. T-stats are reported in small print below the coefficient estimates.

Variables	(1)				(2)			
	64-73	74-84	85-95	96-05	64-73	74-84	85-95	96-05
Intercept	0.82	0.72	1.53	0.99	1.38	4.28	0.65	1.69
	1.36	0.81	2.67	1.83	1.01	3.04	0.82	1.49
<i>Illiq</i>	3.14	0.99	0.22	-0.74	3.30	-0.33	0.51	-0.77
	2.80	2.00	0.36	-1.00	2.78	-0.82	0.88	-1.12
<i>Illiqbeta</i>	-0.04	0.05	-0.02	0.05	-0.04	0.05	-0.02	0.05
	-1.31	1.87	-1.08	3.55	-1.31	1.87	-1.08	3.55
<i>Beta</i>	0.06	0.96	0.09	0.61	0.06	0.96	0.09	0.61
	0.12	1.20	0.19	0.70	0.12	1.20	0.19	0.70
<i>Sdret</i>	-0.20	-0.21	-0.23	-0.15	-0.20	-0.21	-0.23	-0.15
	-1.06	-1.34	-1.58	-0.83	-1.06	-1.34	-1.58	-0.83
<i>Reslnsize</i>	-0.05	-0.28	0.06	-0.05				
	-0.66	-3.66	1.17	-0.65				
<i>Lnsize</i>					-0.05	-0.28	0.06	-0.05
					-0.66	-3.66	1.17	-0.65
<i>Divyld</i>	-0.02	-0.02	-0.01	-0.05	-0.02	-0.02	-0.01	-0.05
	-0.66	-1.08	-0.36	-1.51	-0.66	-1.08	-0.36	-1.51
<i>R100</i>	1.15	0.80	1.26	0.56	1.15	0.80	1.26	0.56
	2.97	1.73	2.88	1.27	2.97	1.73	2.88	1.27
<i>R100yr</i>	0.64	0.54	0.23	0.21	0.64	0.54	0.23	0.21
	2.38	2.36	0.95	0.83	2.38	2.36	0.95	0.83
<i>LnBM</i>	0.10	0.32	0.17	0.14	0.10	0.32	0.17	0.14
	2.07	3.27	2.08	2.14	2.07	3.27	2.08	2.14
<i>RSQ</i>	0.08	0.08	0.06	0.06	0.08	0.08	0.06	0.06
Liquidity Premium (%)	0.19	0.14	0.01	-0.02	0.17	-0.02	0.03	-0.03
T-stat	3.34	2.39	0.20	-0.90	2.76	-0.40	0.82	-1.29

Table 5: Fama-MacBeth Regressions - Two Sub-Periods

The table presents the mean of the coefficients from the monthly cross-sectional regressions of stock returns on the explanatory variables (Eq. (3)). The sub-periods are: 1964-1984 and 1985-2005. *Illiq* is Amihud's (2002) illiquidity measure adjusted for inflation presented in December 2005 prices. *Illiqbeta* is estimated as in Eq. (1). *Beta* is the size sorted portfolio beta calculated every year based on daily observations using the Scholes-Williams (1977) method. *Sdret* is the standard deviation of the stocks returns calculated every year based on daily observations. *Divyld* is the dividend yield. *LnBM* is the logarithm of the book-to-market ratio. *R100* is the stock accumulated return over the last 100 days of the year. *R100yr* is the stock accumulated return from the beginning of the year till the last 100 days. *Lnsize* is the logarithm of the end year market capitalization (in thousands of dollars). *Reslnsize* is the residual of *Lnsize* calculated as in Eq. (2). *RSQ* is the average of the R-Squared of all the monthly regressions in a period. The monthly illiquidity premium is the product of the monthly *Illiq* coefficient and the monthly average *Illiq*. T-stats are reported in small print below the coefficient estimates

Variables	(1)		(2)	
	64-84	85-05	64-84	85-05
Intercept	0.77	1.27	2.90	1.14
	1.41	3.23	2.95	1.68
<i>Illiq</i>	2.01	-0.24	1.40	-0.10
	3.38	-0.50	2.29	-0.23
<i>Illiqbeta</i>	0.01	0.01	0.01	0.01
	0.29	1.21	0.29	1.21
<i>Beta</i>	0.53	0.34	0.53	0.34
	1.11	0.70	1.11	0.70
<i>Sdret</i>	-0.21	-0.19	-0.21	-0.19
	-1.69	-1.67	-1.69	-1.67
<i>Reslnsize</i>	-0.17	0.01		
	-3.14	0.23		
<i>Lnsize</i>			-0.17	0.01
			-3.14	0.23
<i>Divyld</i>	-0.02	-0.03	-0.02	-0.03
	-1.17	-1.49	-1.17	-1.49
<i>R100</i>	0.97	0.93	0.97	0.93
	3.18	2.98	3.18	2.98
<i>R100yr</i>	0.59	0.22	0.59	0.22
	3.36	1.27	3.36	1.27
<i>LnBM</i>	0.21	0.16	0.21	0.16
	3.80	2.95	3.80	2.95
<i>RSQ</i>	0.08	0.06	0.08	0.06
Liquidity Premium (%)	0.17	-0.01	0.07	0.00
T-stat	4.02	-0.35	1.88	-0.05
T-stat of <i>Illiq</i> coefficient differences		2.97	1.99	
T-stat of Liquidity Premium differences		3.72	1.68	

Table 6: Non-parametric Tests for Trend

The table presents the Spearman and Kendall non-parametric tests for trends applied to the 504 monthly estimates obtained from model (3). Columns (1) and (2) apply the non-parametric tests to the coefficient estimates of Illiq, whereas columns (3) and (4) apply the tests to the liquidity premia. The specifications differ in the way the regressions control for size. Columns (2) and (4) use Lnsiz as a control, whereas columns (1) and (3) use Reslnsiz (the component of Lnsiz which is orthogonal to Illiq). Parametric Pearson's correlations are reported for completeness. P-values reported test against the null hypothesis of no trend.

	Coefficients		Liquidity Premium	
	Reslnsiz (1)	Lnsiz (2)	Reslnsiz (3)	Lnsiz (4)
Pearson	-0.154	-0.132	-0.170	-0.120
P-value	0.00	0.00	0.00	0.01
Spearman	-0.127	0.107	-0.163	-0.124
P-value	0.00	0.02	0.00	0.01
Kendall	-0.087	-0.074	-0.108	-0.079
P-value	0.00	0.01	0.00	0.01

Table 7: Monthly Excess Return Averages of Liquidity Based Portfolios

In each year between 1964 and 2005 we sort the stocks into ten deciles based on their previous-year Amihud illiquidity measure (*Illiq*). The top decile consists of the most illiquid stocks and the bottom decile consists of the most liquid stocks. We construct three portfolios: the top decile portfolio assigns equal weights to the most illiquid stocks. The bottom decile portfolio assigns equal weights to the most liquid stocks, and the top-minus-bottom portfolio is long in the top decile and short in the bottom decile portfolios. The portfolios are rebalanced annually. Panel A reports the average monthly excess returns (returns less risk free rate) for each of the three portfolios during the four sub-periods 1964-1973, 1974-1984, 1985-1995, and 1996-2005. Panel B reports the average monthly excess returns over the periods 1964-1984 and 1985-2005.

Panel A

Years	Top Decile		Bottom Decile		Top minus Bottom	
	Excess Return	T-stat	Excess Return	T-stat	Excess Return	T-stat
64-73	0.79%	1.55	0.06%	0.13	0.73%	2.58
74-84	1.63%	3.01	0.17%	0.37	1.47%	3.97
85-95	0.73%	1.70	0.97%	2.62	-0.23%	-0.86
96-05	0.95%	2.34	0.66%	1.56	0.29%	0.79

Panel B

Years	Top Decile		Bottom Decile		Top minus Bottom	
	Excess Return	T-stat	Excess Return	T-stat	Excess Return	T-stat
64-84	1.23%	3.30	0.11%	0.37	1.12%	4.73
85-05	0.84%	2.82	0.82%	2.96	0.01%	0.06
T-stat of Excess Return period differences	0.83		1.70		3.40	

Table 8: Averages of Out-of-Sample Four-Factor Alphas of Liquidity Based Portfolios

In each year between 1964 and 2005 we sort the stocks into ten deciles based on their previous-year Amihud illiquidity measure (*Illiq*). The top decile consists of the most illiquid stocks and the bottom decile consists of the most liquid stocks. We construct three portfolios: the top decile portfolio assigns equal weights to the most illiquid stocks. The bottom decile portfolio assigns equal weights to the most liquid stocks, and the top-minus-bottom portfolio is long in the top decile and short in the bottom decile portfolios. The portfolios are rebalanced annually. Panel A reports the average monthly out-of-sample four-factor alphas calculated using Eq. (4) for each of the four sub-periods: 1964-1973, 1974-1984, 1985-1995, and 1996-2005. Panel B repeats the analysis for two sub-periods: 1964-1984 and 1985-2005.

Panel A

Years	Top Decile		Bottom Decile		Top minus Bottom	
	Alphas	T-stat	Alphas	T-stat	Alphas	T-stat
64-73	0.44%	3.52	-0.28%	-2.49	0.72%	3.74
74-84	0.43%	3.85	0.06%	0.64	0.37%	2.64
85-95	0.09%	0.66	0.08%	1.36	0.01%	0.07
96-05	0.03%	0.16	0.12%	0.95	-0.08%	-0.36

Panel B

Years	Top Decile		Bottom Decile		Top minus Bottom	
	Alphas	T-stat	Alphas	T-stat	Alphas	T-stat
64-84	0.44%	5.23	-0.10%	-1.46	0.54%	4.57
85-05	0.07%	-0.25	0.10%	1.49	-0.03%	-0.25
T-stat of Alpha period differences	2.51		2.09		3.19	

Table 9: Averages of Out-of-Sample Four-Factor Alphas – Portfolios Pre-Sorted by Size

In each month between 1964 and 2005 we sort the stocks in our sample to 5 size quintiles based on the previous end-of-year size. Sizes 1 to 5 refer to the smallest to largest size quintiles. Within each size quintile we sort the stocks into 5 illiquidity quintiles based on the previous year's Amihud illiquidity measure. We then form 5 long-short liquidity-based trading portfolios, one for each size quintile. The portfolios are long in the most illiquid stocks and short in the most liquid stocks within each size quintile. The portfolios are rebalanced annually. Panel A reports the average monthly out-of-sample four-factor alphas calculated using Eq. (4) for each one of the four sub-periods: 1964-1973, 1974-1984, 1985-1995, and 1996-2005. In Panel B the sample period is split into two sub-periods. The panel reports the alphas for these two sub-periods and the T-statistics of the difference between period alphas.

Panel A – Four sub-periods

Years	Size 1		Size 2		Size 3		Size 4		Size 5	
	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat
64-73	0.98%	3.96	0.74%	2.89	0.73%	2.93	0.49%	2.15	0.49%	2.30
74-84	0.34%	1.52	0.49%	1.97	0.39%	1.85	0.05%	0.24	0.07%	0.38
85-95	0.51%	1.95	0.39%	1.76	0.11%	0.55	0.35%	1.96	0.06%	0.46
96-05	0.16%	0.52	-0.22%	-0.74	-0.07%	-0.29	-0.57%	-1.99	-0.14%	-0.69

Panel B – Two sub-periods

Years	Size 1		Size 2		Size 3		Size 4		Size 5	
	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat
64-84	0.64%	3.87	0.61%	3.42	0.55%	3.41	0.26%	1.66	0.27%	1.93
85-05	0.35%	1.70	0.10%	0.52	0.03%	0.17	-0.09%	-0.55	-0.03%	-0.28
T-stat of Alpha period differences	1.13		2.01		2.34		1.54		1.63	

Table 10: Summary Statistics and Cross-Sectional Correlations for Alternative Liquidity Measures

Panel A reports the average of the monthly cross-sectional sample statistics for the alternative liquidity measures and size. In each month during the sample we calculate the mean, median, min, max, and standard deviation of the variables. The table reports the time averages of these statistics. The sample period is 1963-2004. Illiq is Amihud's (2002) illiquidity measure adjusted for inflation presented in December 2005 prices. Turnover is the sum of monthly stock volume values divided by the average number of outstanding shares throughout the year. Lndvol is the average of the logarithm of the daily dollar volume (in thousands of dollars). Lnsiz is the logarithm of the end of year market capitalization (in thousands of dollars). Panel B presents averages of the monthly cross-sectional Pearson's correlations between the liquidity measures and Lnsiz.

Panel A – Summary Statistics

Variable	Mean	Median	Min	Max	Std.
Illiq	0.074	0.024	0.001	1.01	0.13
Turnover	0.54	0.43	0.02	5.14	0.43
Lndvol	12.08	12.11	7.96	15.91	1.55
Lnsiz	12.95	12.88	9.49	17.08	1.43

Panel B - Correlations

Variable	Turnover	Lndvol	Lnsiz
Illiq	-0.19	-0.68	-0.61
Turnover		0.37	-0.03
Lndvol			0.87

Table 11: Fama-MacBeth Regressions Using Turnover and Dollar Volume

The table presents averages of coefficients from monthly cross-sectional regression of stock returns on the explanatory variables (Eq. (3)). Turnover is the sum of monthly stock volume values divided by the average number of outstanding shares throughout the year. Lndvol is the log of the average daily dollar volume (in thousands of dollars). Beta is the size sorted portfolio beta calculated every year based on daily observations using the Scholes-Williams (1977) method. Sdret is the standard deviation of the stocks returns calculated every year based on daily observations. Divyld is the dividend yield. LnBM is logarithm of the book-to-market ratio. R100 is the stock accumulated return over the last 100 days of the year. R100yr is the stock accumulated return from the beginning of the year until the last 100 days. Lnsiz is the logarithm of the end year market capitalization (in thousands of dollars). RSQ is the average of the R-Squared of all the monthly regressions in a period. T-stats are reported in small print below the coefficient estimates.

Variables	(1)		(2)	
	64-84	85-05	64-84	85-05
Intercept	2.92	1.09	1.68	0.81
	3.11	1.65	1.42	0.89
Turnover	-0.82	-0.06		
	-4.18	-0.69		
Lndvol			-0.18	-0.03
			-2.30	-0.53
Beta	0.54	0.43	0.43	0.42
	1.13	0.84	0.91	0.83
Sdret	0.01	-0.17	-0.07	-0.16
	0.10	-1.66	-0.65	-1.69
Lnsiz	-0.18	0.01	-0.02	0.05
	-3.56	0.28	-0.23	0.55
Divyld	-0.02	-0.03	-0.02	-0.03
	-1.06	-1.52	-1.23	-1.51
R100	0.96	0.96	0.86	0.93
	3.19	3.08	2.76	3.02
R100yr	0.69	0.22	0.58	0.21
	3.82	1.26	3.29	1.21
LnBM	0.22	0.18	0.22	0.18
	2.91	2.93	2.98	2.96
RSQ	0.08	0.06	0.08	0.06

Table 12: Out-of-Sample Alphas Using Alternative Liquidity Measures – Portfolios Pre-Sorted by Size

In each month between 1964 and 2005 we sort the stocks in our sample to 5 size quintiles based on the previous end-of-year size. Sizes 1 to 5 refer to the smallest to largest size quintiles. Within each size quintile we sort the stocks into 5 illiquidity quintiles. We then form 5 long-short liquidity-based trading portfolios, one for each size quintile. The portfolios are long in the most illiquid stocks and short in the most liquid stocks within each size quintile. The portfolios are rebalanced annually. The table reports the average monthly out-of-sample four-factor alphas calculated using Eq. (4) for each one of the two sub-periods: 1964-1984 and 1985-2005. The liquidity measure used in Panel A is turnover defined as the sum of monthly stock volume values divided by the average number of outstanding shares throughout the year. The liquidity measure used in Panel B is the average of the daily dollar volume (in millions of dollars).

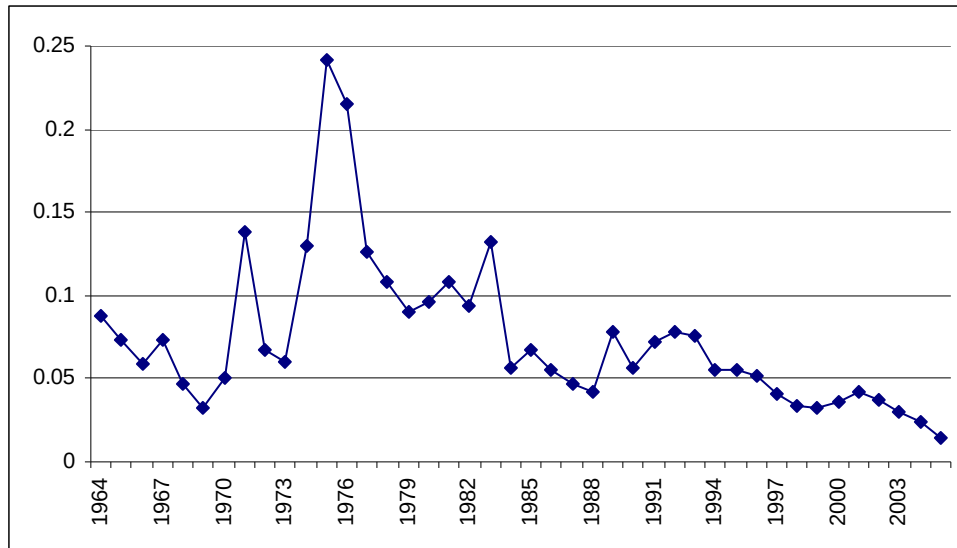
Panel A - Turnover

Years	Size 1		Size 2		Size 3		Size 4		Size 5	
	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat
64-84	0.73%	3.93	0.83%	4.32	0.50%	2.75	0.47%	2.49	0.55%	3.10
85-05	0.66%	3.34	0.33%	1.70	0.27%	1.53	0.04%	0.23	0.01%	0.03
T-stat of Alpha period differences	0.26		1.83		0.90		1.64		2.12	

Panel B – Dollar Volume

Years	Size 1		Size 2		Size 3		Size 4		Size 5	
	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat	Average Alpha	T-stat
64-84	0.81%	4.23	0.83%	4.36	0.57%	3.13	0.42%	2.39	0.45%	2.79
85-05	0.68%	3.27	0.39%	1.99	0.29%	1.61	0.01%	0.08	-0.01%	-0.06
T-stat of Alpha period differences	0.47		1.62		1.10		1.62		2.16	

Figure 1 – Illiq Averages over Time



The figure presents the yearly averages of the monthly cross-sectional Amihud's illiquidity measures (Illiq) scaled for inflation in December 2005 prices.

Figure 2: Average Illiquidity Cross-Sectional Regression Coefficients and Illiquidity Premiums

Figure A – Illiquidity Coefficient

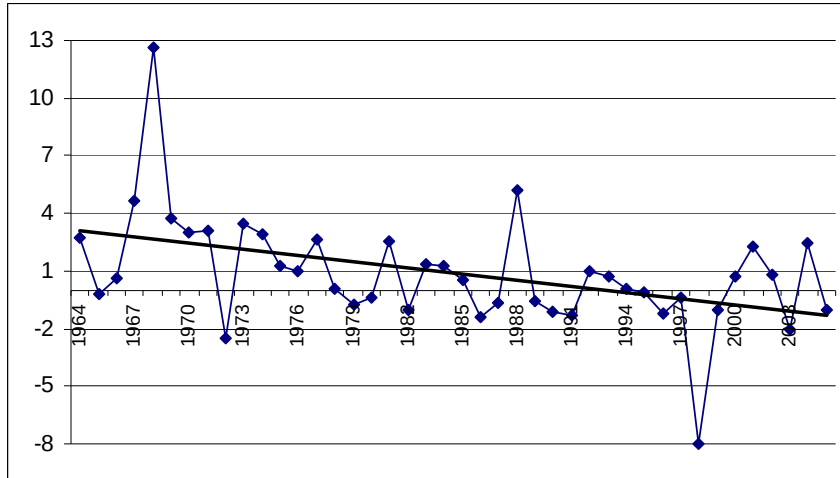


Figure B – Illiquidity Premium

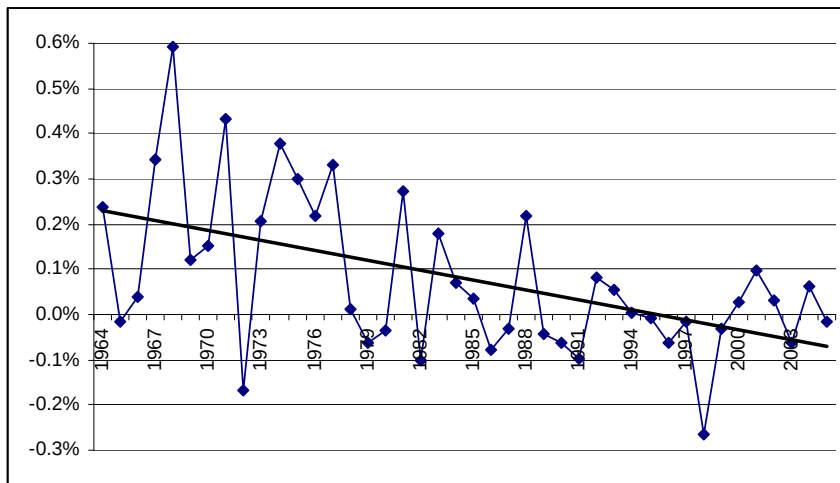
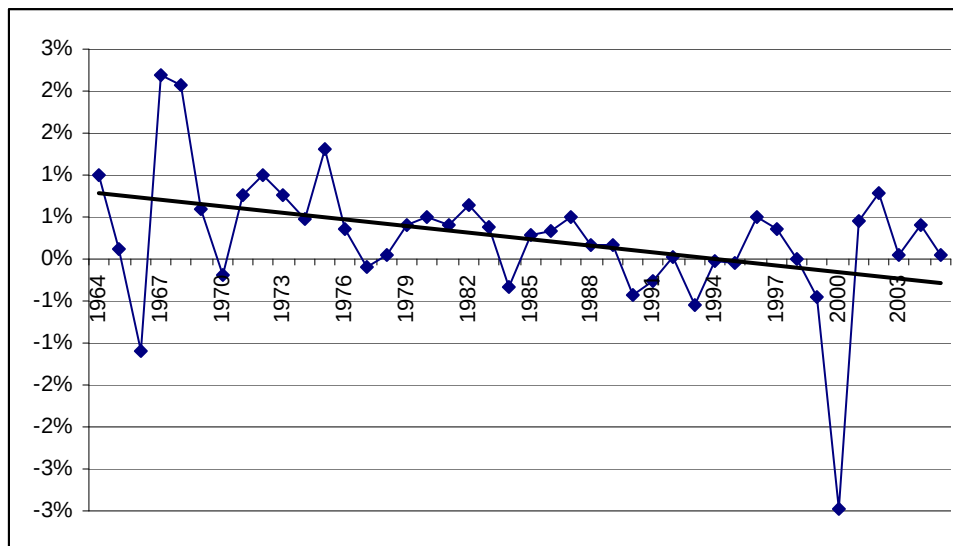


Figure A presents the yearly coefficient averages of Amihud's illiquidity from January 1964 to December 2005 based on specification (1) of table 3. Figure B presents the twelve month average for each year of the Liquidity Premium measure (see Eq.6) from January 1964 to December 2005 based on specification (1) of table 3.

Figure 3: Alpha of Long-Short Liquidity Based Portfolios



The figure presents the twelve month average for each year of the four-factor alphas of a portfolio that is long in the most illiquid stocks and short in the most liquid stocks.